CHANGE OF RING AND PULLBACK OF SHEAVES OF MODULES IS LEFT ADJOINT TO PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sheaves.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Tensor-restrict adjunction for sheaves of modules: Hom_{O}(O tensor_{f^{-1}f_*O} f^{-1}G, F) = Hom_{f^{-1}f_*O}(f^{-1}G, F_{f^{-1}f_*O}). invoked at proof lines 3205-3212. Provides the first equality of the chain.
- Push-pull adjunction for modules: Mor_{Mod(f^{-1}f_*O)}(f^{-1}G, F) = Mor_{Mod(f_*O)}(G, f_*F). invoked at proof lines 3212-3215. Provides the second equality of the chain.

Source reference. Original lemma 008X: sources/stacks-sheaves/pinned-original-source.tex, source lines 3113–3113. Cited at statement lines 3196-3197 to name the module structure on the left-hand side; part of the asserted bijection.

Source reference. Original lemma 008W: sources/stacks-sheaves/pinned-original-source.tex, source lines 3099–3099. Cited at statement lines 3196-3197 to name the f_*O-module structure on f_*F; part of the asserted bijection.

Source reference. Original lemma 008A: sources/stacks-sheaves/pinned-original-source.tex, source lines 2133–2133. Cited at proof lines 3212-3213 as the justification of the first equality; load-bearing inside the counted span.

Source reference. Original lemma 008Y: sources/stacks-sheaves/pinned-original-source.tex, source lines 3150–3150. Cited at proof lines 3213-3215 as the justification of the second equality; load-bearing inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{O}$ be a sheaf of rings on X. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Let $\mathcal{G}$ be a sheaf of $f_*\mathcal{O}$-modules. Then*

$$\mathrm{Mor}_{Mod(\mathcal{O})}(\mathcal{O} \otimes_{f^{-1}f_*\mathcal{O}} f^{-1}\mathcal{G}, \mathcal{F}) = \mathrm{Mor}_{Mod(f_*\mathcal{O})}(\mathcal{G}, f_*\mathcal{F}).$$

Here we use Lemmas 008X and 008W, and we use the canonical map $f^{-1}f_*\mathcal{O} \to \mathcal{O}$ in the definition of the tensor product.

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Proof. This follows from the equalities

$$\begin{aligned} \mathrm{Mor}_{Mod(\mathcal{O})}(\mathcal{O} \otimes_{f^{-1}f_*\mathcal{O}} f^{-1}\mathcal{G}, \mathcal{F}) &= \mathrm{Mor}_{Mod(f^{-1}f_*\mathcal{O})}(f^{-1}\mathcal{G}, \mathcal{F}_{f^{-1}f_*\mathcal{O}}) \\ &= \mathrm{Mor}_{Mod(f_*\mathcal{O})}(\mathcal{G}, f_*\mathcal{F}). \end{aligned}$$

which are a combination of Lemma 008A and 008Y. □